Highlander Health Institute

Centers for Medicare & Medicaid Services
Department of Health and Human Services
Attention: CMS-0042-NC
P.O. Box 8013, Baltimore, MD 21244-8013.

Re: Docket No. CMS-0042-NC "Request for Information; Health Technology Ecosystem"

To the Centers for Medicare & Medicaid Services:

Thank you for publishing the Federal Register notice entitled "Request for Information; Health Technology Ecosystem" (CMS-0042-NC) (the "Notice"). Highlander Health appreciates the Agency's commitment to advancing health technology infrastructure and improving data-based decision-making. Highlander Health Institute ("HHI" or the "Institute") is a public interest platform that develops learning labs in a variety of settings. The Institute brings together provider, regulatory, academic, tech and life sciences collaborators to set the standard in next-generation evidence generation and care. Our aim is to nurture pressure-tested, practical, and widely shared answers to longstanding problems in how we assess and access treatments

The Institute greatly appreciates the efforts of the Centers for Medicare & Medicaid Services ("CMS") to advance health data interoperability and improve the ease with which clinical and real-world data ("RWD") can be collected, curated, and analyzed for use in regulatory decision-making. We welcome the opportunity to provide feedback on the state of data interoperability and broader health technology infrastructure. This comment letter is organized to directly respond to specific questions from Section E of the RFI (Technology Vendors, Data Providers, and Networks).

As we move toward a more interoperable national healthcare infrastructure, it is important to adopt standards and schema that enable harmonization of data from clinical investigations and data derived from real-world sources. One of Highlander Health Institute's main theses is the importance of thinking about clinical trial data and RWD as two sides of the same coin - they are both examples of clinical data. The only difference between the two are variations in the circumstances and controls through which the data are collected. For this comment letter, we define RWD as clinically relevant data collected outside the boundaries of a formal clinical trial. Data sources include EHRs, administrative claims, -omics data, imaging data, digital pathology, and other formats. The value in considering clinical trial and RWD as differing versions of clinical data is that it provides a more holistic perspective of a patient's health and the impact of a medical intervention. Clinical trials are typically a point-in-time snapshot that assesses whether a medical product works in a controlled environment. RWD provides information about how a medical product works in a broader context, which can include social determinants of health and other factors that may impact wellness. A full picture of a patient's health can only be painted when you leverage both data sources. Our comments are aimed toward achieving this unified vision of patient

health; this vision for a modern data infrastructure for clinical research is outlined in a 2024 paper in the *Journal of the American Medical Association*, with the senior author as cofounder of Highlander Health.[1]

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

Access to broader, more timely, and more complete sources of data would improve research related to value-based care, a key priority for CMS. Specific sources of valuable data include real-time claims data, Medicaid claims data, Medicare Advantage encounter data, and data related to social determinants of health. We understand there are significant barriers to providing these types of data, especially real-time data and unified Medicaid data, but believe they are important to advancing patient care and CMS's strategic priorities.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

One of the most impactful application programming interfaces ("API") would be an API that enables the ability to combine payor data with clinical data. Such an API and the resulting use cases would greatly improve the quality and ease of completing reimbursement efforts. A pilot amenable to such an API would involve the Cell and Gene Therapy ("CGT)" Access Model.[2] Enabling the accessibility of outcomes-based data from the Access Model would improve patient care in a number of ways. First it would continuously improve the CGT Access Model Program by enabling shared learnings. Second, such data could be shared with the Food and Drug Administration ("FDA") to improve long-term follow up of CGT safety and effectiveness. One of the major unknowns with these products is how they perform over a 15+ year time horizon. Sharing data with other regulators would enable better stewardship of public health and improve how patients are treated.

Another potential API would be an interface pertaining to specific coverage with evidence development ("CED") programs, such as the Transcatheter Aortic Valve Replacement CED program.[3] Enabling access to real-time data from CED programs would lead to faster innovation, particularly in terms of refining new and developing technologies and identifying unmet patient needs. Additionally, such data access would enable more transparent and equitable coverage decisions through broad access and the facilitation of third-party validation of needed analyses. Finally, API access to CED data would lead to improved clinical decision-making and create a feedback loop that improves both the program design and underlying policy related to the CED decision.

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[1] Franklin, et al., "Modernizing the Data Infrastructure for Clinical Research to Meet Evolving Demands for Evidence." Available here.
[2] Centers for Medicare and Medicaid Services, "Cell and Gene Therapy (CGT) Access Model." Available here.
[3] CMS, "Transcatheter Aortic Valve Replacement." Available here.

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

We define proprietary APIs as those that do not conform to open standards, meaning they are built by vendors, use non-standard payload formats, and are not typically publicly documented or accessible. Promoting the use of open APIs encourages transparency and accountability and allows for stronger interoperability in the health IT ecosystem. There are a few options to encourage the use of open, standards-based, publicly available APIs over proprietary APIs. One approach would be to tie CMS program eligibility to the use of certain APIs, such as HL7 Bulk FHIR Access. If EHRs are structured to accommodate Bulk FHIR Access, the resulting positive impacts on effective value-based care, clinical trial recruitment, and public health monitoring would be sizable.[4] However, this move towards open, standards-based APIs would need to be accompanied by change management efforts. Organizations would need to publish FHIR implementation guides as a condition of CMS program participation in order to generate wide-scale learnings.

A middle ground option could be to partner with the Office of the National Coordinator ("ONC") to continue to certify and endorse standards-based APIs. Working with an eye towards usability, provenance, and interoperability as guiding principles for certification, CMS and ONC could promote and encourage the use of publicly available APIs and fund API development by providing grants or cooperative agreements. A specific example could involve developing FHIR implementation guides for key topics of interest, such as social determinants of health or post-acute care.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide provider directory of FHIR endpoints would improve access to health information for patients by simplifying third party health app connections and supporting record integration from multiple systems. These improvements would have the downstream effect of helping address health disparities by providing a single point-of-access tool for low-resource patients or safety-net providers. Such a directory would also improve access for providers by improving the speed of referrals and care coordination, reducing administrative burden, and enabling the use of certain clinical decision support tools that require multi-source patient data. Finally, this directory would improve payer access by improving member data retrieval, supporting interoperability rules, and encouraging innovation in plan-sponsored apps.

The most effective way to create and maintain such a directory would likely be through a public-private partnership. ONC, given its role as a standard-setting organization, could "own" the directory and

[4] *See* EHR Good Neighbor, "Case Studies." Available here.

partner with organizations such as HL7 and state health information exchanges to create a routinely-updated directory with searchable endpoint metadata and validated endpoint functionality. Maintenance of such a directory could be funded through TEFCA infrastructure grants or fee structures for vendors submitting bulk updates.

TD-6. What unique interoperability functions does TEFCA perform?

TEFCA enables cross-network record location and retrievals across a wide array of organizations. TEFCA's legal and operational rules for participants enable this ability, as common privacy, security, consent, identity proofing, and data handling standards are crucial enablers of interoperability. TEFCA is best thought of as complementary to other health IT interoperability efforts, including FHIR APIs. Highlander does not recommend CMS adopting an alternative to TEFCA at this time.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

USCDI has been instrumental in creating a baseline for structured, interoperable health information exchange. It has created a minimum dataset for nationwide exchange and enabled a shared vocabulary across various organizations. Continuing to promote standardization of key clinical and demographic data will improve the consistency or record aggregation and analytics that USCDI has already enabled.

a. Does it contain the full extent of data elements you need?

USCDI contains key limitations as it relates to real-world data interoperability. Improvements are needed for specific types of data, including claims data, genomic data, social determinants of health, and device and sensor data. Additionally, USCDI could be improved by accommodating provenance and metadata for source data and system of origin.

Further, USCDI represents "core" data elements, meaning that it lacks the needed specifics for care/disease specialties. Efforts like USCDI+ (for cancer) are beginning to address this, but USCDI should be thought of as a starting point, rather than a finished product. For example, at the May 2024 Cancer Research Data Exchange Summit[5], a presentation discussed identification of immune-related adverse events. USCDI itself does not contain the needed set of data elements for identifying many such adverse events, pointing to a clear need for disease area-specific augmentations. There is complementary work ongoing in codeX[6], a HL7 FHIR accelerator. mCODE[7] and CardX[8] should serve as models for specialty versions and next-generation versions of USCDI data elements.

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

[5] USCDI+ Cancer Post-Summit Summary. Available here.
[6] CodeX, available here.
[7] HL7 International, "minimal Common Oncology Data Elements (mCODE) Implementation Guide." Available here.
[8] HL7 International, "CardX Hypertension Management Use Case." Available here.

USCDI's limitations are due to both the USCDI format and the way it is utilized. As it relates to format, USCDI defines data classes and elements but does not always define a standardized structure for those categories. As mentioned earlier, USCDI as currently constructed does not always reflect data provenance, quality, or contextual relationships. As it pertains to utilization, certified health IT developers and vendors often treat USCDI as a compliance floor, rather than a tool to create deeper interoperability. Additionally, variability in USCDI implementation can lead to inconsistencies in data capture.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

Adding more data elements to USCDI would add value but come with scoping challenges. Increasing the number of data elements in USCDI would create more granular data that could improve public health surveillance, cross-sector care, and population health activities. However, adding data elements without clear implementation guidance or thoughtfulness about which data elements are truly necessary may lead to an undue burden on health care providers. If USCDI was to be expanded, it would be important to better understand how automated data capture, patient self-reporting, and AI-based solutions, including natural language processing ("NLP"), could ease the data collection burden that currently falls on providers.

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

Highlander Health supports standards-based interoperability as foundational, but we recognize that the real world is messy: not all clinically meaningful data fits neatly into rigid data schemas like USCDI. Improvements in large language models ("LLMs") and NLP have created a window of opportunity to balance structure and flexibility in health data exchange. The benefits of allowing a less structured format likely include improved data coverage and reduced documentation burden for clinicians.[9] However, necessary guardrails must be in place. Any format must be non-proprietary to enable open access and prevent vendor lock-in. Additionally, all NLP or LLM transformations need to be auditable and transparent. This is easier said than done, as any LLM would have billions of parameters. This naturally points to the need for robust validation and quality control standards for AI tools in this arena. Providing reference models or benchmarks for extracting structured USCDI-like elements from unstructured data would be enormously beneficial. Highlander Health recommends that CMS collaborate with FDA and the National Institutes of Health to create a harmonized approach to validation. Such an approach would include uniform guideposts to developing, validating, and continuously monitoring algorithms regardless of eventual use case.

[9] See Phelan et al., "Beyond compliance with the 21st Century Cures Act Rule: a patient controlled electronic health information export application programming interface" Available here.

In short, we believe CMS and ONC should embrace a dual-path approach: (1) maintain and evolve USCDI as the backbone for structured data exchange, while (2) simultaneously encouraging safe, transparent use of AI models to enhance data completeness and utility from less structured inputs.

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

Shifting the focus of certification from internal software functionality to API-enabled capabilities would better align the ONC Health IT Certification Program with real-world health data use, modern interoperability needs, and emerging innovation ecosystems. API-focused certification would support ecosystem flexibility and encourage innovation by third parties by enabling a marketplace of modular tools for care coordination, analytics, public health, and research. Prioritizing API-enabled capabilities would also align with modern technical standards, reduce vendor lock-in, and support policy goals related to value-based care by improving scalable, standards-based health information exchange.

Another benefit is that the transition to API-enabled capabilities dovetails with the evolution of the technology industry. Early incarnations of tech systems were monolithic, meaning that all components were built as a tightly-integrated unit. Over time, this transitioned to microservices, in which individual components are decoupled and interact through standard APIs. Bringing the now-common microservices approach to healthcare IT would result in quicker adoption of new technology and capabilities.

Finally, moving to API-enabled capabilities would improve FDA and CMS alignment. FDA's recent comment opportunities related to Dataset-JSON exchange formats and HL7/FHIR show that it has similar goals to CMS: being able to use and reuse data and IT infrastructure to accomplish each agency's responsibilities and improve patient care. As mentioned earlier, the CGT use case to improve both the Access Model and FDA's need for long-term followup data would be a shining example of how API-enabled capabilities could serve as a rising tide that lifts all boats. Another example would be using this opportunity to construct a set registration endpoints from FHIR. In other words, what does a set of FHIR Conditions, Observations, Encounters, Procedures, deceasedDateTime that feed into major adverse cardiovascular events look like? These endpoints could then be aligned with CMS quality measures to improve value-based care applications.

b. What would be the drawbacks?

Potential drawbacks include an increased testing and maintenance burden on vendors and health systems. For example, API-first certification may require more frequent updates and regression testing as APIs evolve and new data classes (e.g., USCDI+ domains) are added. Additionally, a focus on API

certification must be accompanied by implementation guidance and auditing to prevent inconsistent real-world implementation.

Highlander recommends evolving the certification paradigm to treat API-enabled capabilities as the primary certification focus, while maintaining a baseline for clinical functionality that ensures patient safety and provider usability and supporting robust post-certification surveillance to assess real-world utility of certified APIs.

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

Yes, CMS should endorse non-CMS data sources and networks to create a more comprehensive, efficient, and interoperable national health data infrastructure. Non-CMS networks, such as qualified health information networks (QHINs), payer data platforms, clinical research networks, social determinants of health data collaboratives, and commercial health data utilities, often hold critical longitudinal information across care settings, payers, and populations. Endorsing these data sources would advance CMS' mission to improve value-based care, health equity, and regulatory-grade RWD development.

To maximize trust, transparency, and utility, CMS endorsement should be based on whether a data source or network meets key technical, operational, and ethical benchmarks. Potential endorsement criteria should include standards-based interoperability, data quality and completeness, and conformance with key privacy and security requirements. Additionally, endorsed data sources should enable broad patient and provider access.

CMS endorsement of non-CMS data networks, if rooted in open standards, transparency, and performance, would accelerate a future where trusted data flows to where it's needed most: at the point of care, for public health, and for patients themselves.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

Enabling API-based access to the full scope of electronic health information (EHI) and going beyond USCDI opens the door to transformational use cases that advance patient care, health system performance, and other activities, such as medical product discovery.[10] If implemented effectively, access to the entirety of a patient's EHI would catalyze a more intelligent, responsive, and patient-centered health system. New opportunities could include enabling true longitudinal care by giving health care providers access to a patient's full clinical context over time. Additionally, patients could reap the benefits of true data portability by being able to easily consolidate and share their entire health record.

[10] See Phelan et al., "Beyond compliance with the 21st Century Cures Act Rule: a patient controlled electronic health information export application programming interface" Available here.

Finally, as treatment modalities improve and move towards precision medicine, API advancement combined with access to full patient EHI would create richer datasets that enable and refine these treatments.

APIs represent the infrastructure layer for true health IT interoperability. By removing artificial constraints on what data flows and enabling access to a patient's full EHI, CMS can enable a learning health system that is both technically interoperable and clinically meaningful. Again, this step towards interoperability would enable cell and gene therapy long-term followup and CED project goals.

a. What are the primary obstacles to this?

The primary obstacle to APIs providing access to the entirety of a patient's EHI is a lack of incentive for the standardization for non-USCDI data. USCDI defines core structured elements, but most EHI (e.g., imaging metadata, pathology PDFs, clinical notes) lacks consistent FHIR representation. This hinders interoperability and requires burdensome processing by systems receiving the data. Another obstacle includes limited market incentives to support full EHI access. Payers, providers, and vendors likely would need demand-side incentives in order to enable this approach and develop the associated infrastructure needed to make such a policy directive impactful.

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

Modern technology advancements, particularly LLMs and NLP, can help address the traditional limitations associated with unstructured data. Accordingly, access to full EHI is a more realistic proposition than ever before, but we must move towards this goal with a realistic understanding of America's health system and current capabilities. We support maintaining USCDI as a baseline for plug-and-play interoperability, but enabling full-EHI access through certified APIs is essential for the next generation of data-driven health innovation. CMS should continue to expand USCDI and USCDI+ for high-value domains (e.g., oncology, cardiology, social needs) and begin to think through tooling and guidance (e.g., NLP libraries, reference implementations) to help health systems process unstructured data responsibly and eventually enable full EHI access.

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

Increased use of bulk FHIR would improve use cases and data flow by improving the efficiency of population-level data access. Bulk FHIR enables authorized users (e.g., payers, health systems, researchers) to retrieve large datasets for many patients in one request, rather than making individual patient API calls. This supports use cases like risk stratification, quality measurement, outcomes

research, and public health reporting. Bulk FHIR would also provide better support for value-based care by reducing the complexity of aggregating data from many providers or EHRs.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

Despite substantial technical progress, healthcare data sharing networks still face major sustainability challenges, including high operational costs, insufficient incentive alignment, and uneven participation. CMS and ASTP/ONC have a unique role to play in ensuring these networks deliver measurable value. First, CMS could condition participation in value-based care models, accountable care organizations, Medicare Advantage plans, and other reimbursement streams on active network participation (e.g., TEFCA, QHINs), ensuring a reliable base of supply and demand. Leaning on its authority to set standards and certifications, ASTP/ONC should ensure that shared services that enable health IT interoperability such as endpoint directories and FHIR implementation guides are funded and maintained.

Health data networks should be treated as critical national infrastructure, similar to highways. CMS and ONC have a joint responsibility to scale trusted digital highways and ensure the networks serve patients, not just platforms.

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Highlander Health looks forward to collaborating with CMS to advance the nation's health IT infrastructure. Please contact Nicholaas Honig at Nickh@highlanderhealth.com with any questions regarding these comments or other issues related to evidence generation policy and infrastructure.

Sincerely,

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